Accelerating Mars Landing Site Selection Through Predictive Geophysical Modeling

Arv Jain

Abstract

Selecting sites for planetary exploration is a difficult task, as changing regions, outdated planetary scans, the vast scope of the Martian surface, and low-resolution data complicate

decision-making, and scanning optimal sites remains resource-intensive and often wasteful due to the large number of locations scanned and unused. The project aims to make initial site selections for Martian exploration more efficient and streamlined than current landing site evaluations used for space exploration. Current methods often use rover scanning to predict sub-surface qualities such as water concentration. These methods are slow, often inaccurate, and produce low-quality data. This study uses a machine learning-oriented approach, utilizing low-level surface qualities of Mars, such as topography and mineral densities, to predict deeper geophysical qualities at higher resolutions. The project uses multiple feed-forward MLP neural networks, implemented through Pytorch, to predict various geophysical qualities of the Martian surface. Input data was scaled using RobustScaler to reduce sensitivity to extreme values inherent in noisy Martian datasets, and partitioned via a stratified spatial split across latitude bands to ensure the models were evaluated on geographically distinct regions, preventing the data leakage that would arise from a naive random split. Models underwent hyperparameter tuning through Bayesian Optimization using Optuna's TPE model with R^2 score as the objective, each tuned separately for better accuracy. Models were trained up to 500 epochs with early stopping. Outputs were compared to historical landing sites for validation. Preliminary results indicate that certain geophysical properties, such as water-equivalent hydrogen and gravitational anomalies, can be predicted with high accuracy, with five of six models achieving strong to exceptional performance (Average $R^2 = 0.927$, excluding TI Variance). The pipeline provides a scalable and systematic approach for evaluating candidate Mars landing sites, reducing the need for exhaustive scanning and reliance on low-resolution data.

Introduction and Literature Review

Traditional Martian Landing-Site Selection

Selecting suitable landing sites on Mars is a critical component of planetary exploration. Landing-site selection must balance engineering constraints, such as terrain safety and atmospheric conditions, with scientific objectives, such as investigating Martian geological history and identifying evidence of past water activity. Consequently, selecting a landing site requires extensive analysis of the Martian surface and its underlying geophysical properties. This literature review examines traditional landing-site selection methods, developments in automated geospatial analysis, and previous applications of machine learning to Martian surface and geophysical modeling.

Traditionally, space agencies such as the European Space Agency (ESA) and the National Aeronautics and Space Administration (NASA) have relied on extensive scientific assessments and collaborative workshops to select landing sites. For the ExoMars mission, the European Space Agency outlined several important landing requirements, including low elevation to provide sufficient atmospheric braking and the absence of dangerous surface features such as steep slopes, large rocks, and numerous craters (European Space Agency). The selection process began in December 2013, requiring approximately one year to narrow the proposed landing sites to four candidates and additional time to finalize the selection. A similar approach was used by NASA during the selection of landing sites for the Mars Exploration Rovers. Golombek et al. reported that the selection of Meridiani Planum and Gusev Crater required more than two years, during which approximately 155 potential landing sites were reduced to four finalists based on scientific and engineering considerations. Important evaluation criteria included surface slope,

rock abundance, thermal inertia, dust, and scientific potential (Golombek et al.). These examples demonstrate the extensive analysis required to identify suitable landing sites. However, the length of the selection process does not necessarily indicate inefficiency, as detailed evaluations are essential to ensuring mission safety and scientific value. Instead, the challenge lies in efficiently narrowing a large geographic search space into a smaller set of promising regions. A preliminary screening framework capable of estimating relevant environmental and geophysical properties across broad areas could help researchers prioritize locations for subsequent high-resolution analysis.

Geospatial Analysis and Multi-Criteria Site Evaluation

Advances in Geographic Information Systems (GIS) have allowed researchers to automate portions of planetary site evaluation. GIS enables the integration of multiple geographic datasets into a unified spatial framework, allowing researchers to assess scientific and engineering criteria simultaneously. Pajola et al. (2019) described the importance of GIS-based mapping in planetary landing-site selection, emphasizing its role in evaluating engineering constraints and scientific requirements. Their work examined Martian landing-site proposals for both the ESA ExoMars and NASA Mars 2020 missions, demonstrating how geographic datasets support systematic assessments of candidate regions. More recently, Mashal et al. (2025) developed a Multi-Criteria Overlay Analysis (MCOA) framework to identify preferred exploration zones on Mars. Their approach integrated geological, mineralogical, topographical, and resource-related datasets using GIS. Individual criteria were standardized and combined into composite suitability maps, with engineering constraints used to exclude unsuitable regions. The researchers identified 23 particularly high-scoring regions that merited further investigation. This study demonstrates that automated geospatial analysis can substantially narrow the search space for future Martian

exploration. However, the method remains dependent on the availability, coverage, and spatial resolution of existing geographic datasets. Mashal et al. specifically acknowledged that interpolating lower-resolution datasets onto finer grids could introduce misleading artifacts. Thus, although multi-criteria analysis provides an effective method for integrating known geographic information, it does not inherently estimate properties that are unavailable or insufficiently resolved in the underlying datasets. Another approach was developed by Zhu et al. (2025), who investigated Martian habitat suitability using environmental factors including temperature, terrain, dust activity, water ice, and radiation. Their framework integrated multiple geographic datasets into suitability maps for potential human habitation, identifying promising regions such as southern Arcadia Planitia and southern Utopia Planitia. Although focused on long-term human settlement rather than robotic landing-site selection, this study further demonstrates the potential of integrated geographic analysis to support preliminary assessments of Martian environments.

Together, these studies establish the value of multi-criteria geospatial analysis in identifying promising regions. However, they also highlight the importance of obtaining reliable estimates of the environmental and geophysical properties used in such evaluations. This motivates the investigation of predictive models that can supplement existing geographic datasets.

Machine Learning in Martian Surface and Geophysical Analysis

Machine learning has emerged as another approach to accelerating the interpretation of Martian observational data. Rather than relying exclusively on manual examination or direct physical modeling, machine-learning algorithms can identify relationships within existing datasets and use those relationships to make predictions about previously unseen observations. Meng et al. (2024) developed a Convolutional Neural Network (CNN) to identify geomorphological

similarities between Martian and terrestrial landscapes. By analyzing landforms associated with geological processes, the researchers demonstrated how automated image interpretation could support the identification of scientifically interesting Martian regions. Their work was particularly relevant to identifying features associated with past water activity. However, the approach primarily focuses on geomorphological identification rather than predicting quantitative environmental or geophysical properties. Similarly, Barrett et al. (2022) developed NOAH-H, a deep-learning framework that classifies high-resolution HiRISE imagery into terrain categories. This approach enables automated interpretation of complex surface imagery and can support the identification of potentially hazardous terrain. However, Barrett et al. acknowledged that terrain classification alone cannot definitively determine whether an area is hazardous without additional topographical information and mission-specific engineering constraints.

While these studies demonstrate the effectiveness of machine learning for interpreting Martian surface features, other researchers have investigated its application to quantitative geophysical prediction. Song et al. (2023) developed a machine-learning approach to estimate Martian thermal inertia, a physical property describing how effectively surface materials resist temperature changes. Using data generated from established thermal models, the researchers trained random forest regression models to predict thermal inertia from surface kinetic temperature, latitude, time of day, and other accessible parameters. Their results demonstrated agreement with previous thermal-inertia retrievals and showed that machine learning could rapidly estimate a quantitative Martian surface property.

Unlike image classification approaches, the work of Song et al. demonstrates the potential of machine learning to approximate relationships between measurable environmental variables and geophysical properties. Such models can serve as computationally efficient alternatives to

repeatedly applying more complex physical retrieval procedures. However, their study focuses specifically on thermal-inertia estimation rather than integrating predictions of multiple distinct properties into a broader landing-site evaluation framework. These developments demonstrate two complementary applications of machine learning in Martian exploration: automated identification of surface features and quantitative prediction of planetary properties. Both approaches have the potential to support preliminary site assessment, although their outputs must be interpreted within the limitations of their training data and underlying physical assumptions.

Research Gap

Existing research demonstrates that GIS-based multi-criteria analysis can identify promising Martian exploration regions, while machine-learning models can automate terrain interpretation and estimate quantitative geophysical properties. However, the reviewed studies primarily investigate these capabilities separately. Multi-criteria frameworks such as those developed by Mashal et al. and Zhu et al. integrate existing geographic datasets, while machine-learning studies such as those by Meng et al., Barrett et al., and Song et al. focus on particular classification or prediction tasks. This distinction motivates the investigation of a framework that combines multiple machine-learning-based property predictions with integrated geographic suitability analysis. In particular, estimating several environmental and geophysical properties from available surface observations could support preliminary screening across large geographic regions, allowing researchers to prioritize promising locations for more detailed investigation. Additionally, predictions generated by machine-learning models contain uncertainty, particularly when applied to geographic regions that differ from their training data. Incorporating ensemble-based measures of model disagreement could provide an additional indication of

where predictions may require greater scrutiny, although such disagreement does not necessarily represent calibrated predictive uncertainty.

The Mars Geophysical Prediction Framework (MGPF) investigates this integration by using multiple neural-network models to estimate Martian properties, including water-equivalent hydrogen, dust, surface temperature, temperature variation, gravity anomalies, and thermal-inertia variation. These predictions are combined within an interactive, adjustable suitability framework intended to support preliminary geographic screening. Rather than replacing established landing-site selection procedures, MGPF explores whether machine-learning surrogate models can supplement existing methods by providing rapid, multi-property estimates across a planetary-scale grid. Its objective is to support the prioritization of candidate regions for further scientific and engineering assessment, thereby investigating the potential of predictive modeling to improve the efficiency of early-stage Martian landing-site exploration.

Methodology

Input Features

Input features consisted of surface-level Martian features as well as mineral abundance data. Data quality was set to 2ppd to streamline training. Additionally, the highest quality for the WEH% data was 2ppd, and so the dataset was downscaled to ensure resolution consistency. Data was extracted from JMARS 5 via the polygon extraction tool. Polygons were made over the whole Martian surface, and CSV data was extracted from the tool at 2ppd. The list of features and targets is shown on Table 1.

Preprocessing

Data was extensively preprocessed to allow networks to converge better. First, Latitude and Aspect features, if used, were encoded with sine and cosine to preserve periodicity in the model. Next, rows with missing values were removed to avoid imputing data, which could result in network biases and inaccuracies. Since Martian data is extremely noisy, features were scaled using RobustScaler, reducing sensitivity to extreme values. Lastly, data was split via latitude-bands to ensure the model was evaluated on distinct geographies. This is more appropriate than a random split, which could result in training on geographically similar data while evaluated on distinct regions (overfitting). The spatial split was also stratified to counteract the effects of heterogeneous latitude bins, which would prevent the model from capturing patterns across regions.

Some models like Temperature Variation and TI Variance featured an outlier clipping step early on, as those variables had large outliers decreasing the performance of the network.

Model Architecture and Training

For each target, a separate feed-forward MLP model was constructed, each with its unique hyperparameters. Parameters were optimized via Bayesian Optimization using Optuna's Tree-Structured Parzen Estimator (TPE). R^2 was set as the objective function. The search grid was shared among models, but each was tuned individually to ensure the best-performing parameters per model.

Final training used an ensemble of 5 MLP models to improve stability and reduce variance, aggregated via their mean. Models were trained up to 500 epochs with early stopping at patience

30. Models were trained on the spatially split training set. Mean squared error was set as the loss function.

Outputs

For each model, the mean prediction was recorded along with the standard deviation. Standard deviation served as the uncertainty estimate, with higher uncertainties being penalized heavily in the heatmap. Final performance was evaluated on R^2 score on the spatially split test set.

Heatmap

Features at 8ppd were extracted from JMARS 5 as .tif files, then converted into a CSV. We ran model predictions on each data point on the dataset, creating a 8ppd equivalent for the targets using ML. Points were computed to create a landing suitability map. Each point was normalized between one and zero, and depending on the target variable, was promoted or penalized (See Table 2). Landing goodness heatmaps are projected onto the main site through an API, and users can adjust the weights of targets depending on mission parameters.

Table 1: ***Model Targets and Input Features***

Model	Target	Features
Dust Predictor	OMEGA Ferric/Dust 860nm ratio	MOLA 128ppd Elevation • MOLA 128ppd Slope • MOLA 128ppd Aspect • OMEGA Albedo 1080nm • Dayside Thermal Inertia

Gravity Predictor	Mars Bouguer Gravity Anomaly	MOLA 128ppd Elevation • MOLA 128ppd Slope • MOLA Roughness 2.4km • Dayside TI • Nightside TI • TES Basalt Abundance • TES Rock Abundance • Latitude (N)
Surface Temp	Yearly Avg Surface Temp (C)	Latitude (N) • MOLA 128ppd Elevation • MOLA Roughness 0.6km • MOLA Roughness 2.4km • OMEGA Albedo 1080nm • Dayside TI • Nightside TI
Water Predictor	Neutron Spectrometer % WEH	Latitude (N) • MOLA 128ppd Elevation • OMEGA Albedo 1080nm • Dayside TI • TES Basalt Abundance • Mars Crustal Thickness
Temp Variation	Yearly Surface Temp Variation	Latitude (N) • MOLA Elevation, Slope, Aspect • MOLA Roughness 0.6km • OMEGA Albedo 1080nm • Dayside TI, Nightside TI
TI Variance	TI_Variance (Diurnal Swing)	Latitude (N) • MOLA Elevation, Slope, Aspect • MOLA Roughness 0.6km, 2.4km • TES Basalt, Andesite, Rock Abundance • OMEGA Albedo 1080nm • Mars Crustal Thickness

Note: MOLA = Mars Orbiter Laser Altimeter; OMEGA = Observatoire pour la Minéralogie, l'Eau, les Glaces et l'Activité; TI = Thermal Inertia; TES = Thermal Emission Spectrometer; WEH = Water-Equivalent Hydrogen.

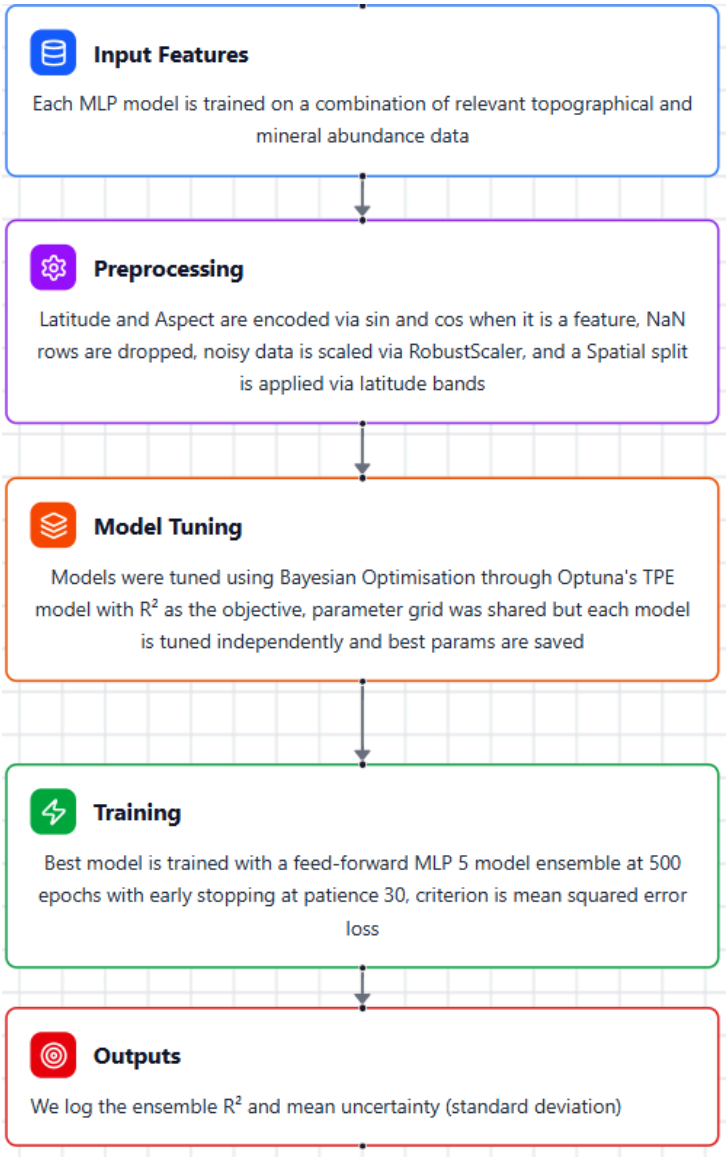
Table 2: ***Target Weights for Heatmap***

Target Feature	Transform / Curve	Boost or Penalty	Effect on Landing Score
Water	tanh(val × 1.8)	Boost	Rapidly increases score at low–medium values, saturates at high values
Dust	1 − val^1.5	Penalty	Low dust gives high score; high dust sharply reduces suitability
Surface Temp	Asymmetric Gaussian around 0.65	Optimal Range Boost	Best near ideal temperature; too cold or hot reduces score
TempVariation	1 − clip(val × 1.8)	Penalty	High temperature swings reduce score
TI Variance	1 − clip(val × 1.8)	Penalty	Highly variable terrain lowers suitability
Gravity	Bell curve centered at 0.5	Optimal-range boost	Extreme gravity values penalized; moderate gravity favored

Table 3: ***Model Architecture and Hyperparameters***

Model	Layers	Units per Layer	Activation	Learning Rate (Opt)	Weight Decay
Water	6	224	Gelu	0.00044	4.48e-05
Dust	4	208	Gelu	0.00018	2.69e-06
Surface Temp	5	240	Gelu	0.00049	5.66e-06
Temp Var	6	240	ELU	0.00044	1.44e-04
Gravity	6	176	Gelu	0.00049	1.35e-05
TI Variance	6	224	Gelu	0.00039	7.21e-05

Figure 1: ***Model Architecture for Landing Site Assessments***



Results

Model Performance and Metrics

MGPF demonstrated high predictive accuracy across most targets, with an average R^2 of 0.927 across five of six models, showing that the models capture meaningful patterns even with noisy, low-resolution data. Surface Temperature and Temperature Variation predictions performed the strongest ($R^2 = 0.9976$ and 0.9808 respectively), while TI Variance remained the only limited model ($R^2 = 0.3416$), attributable to the inherent noise of the derived diurnal swing target rather than pipeline failure. Mean prediction uncertainties reflected these trends, with higher σ values in models with noisier targets. See Table 4 for the full breakdown of scores.

Heatmap Validation

MGPF's landing suitability heatmaps were compared against historical landing sites. Gale Crater and Jezero Crater were consistent with their selection criteria for the Curiosity and Perseverance missions. High-scoring regions corresponded to water-rich areas, reflecting the scientific criteria historically used for site selection. The heatmaps also highlighted additional suitable regions, suggesting MGPF could identify promising sites beyond those historically chosen.

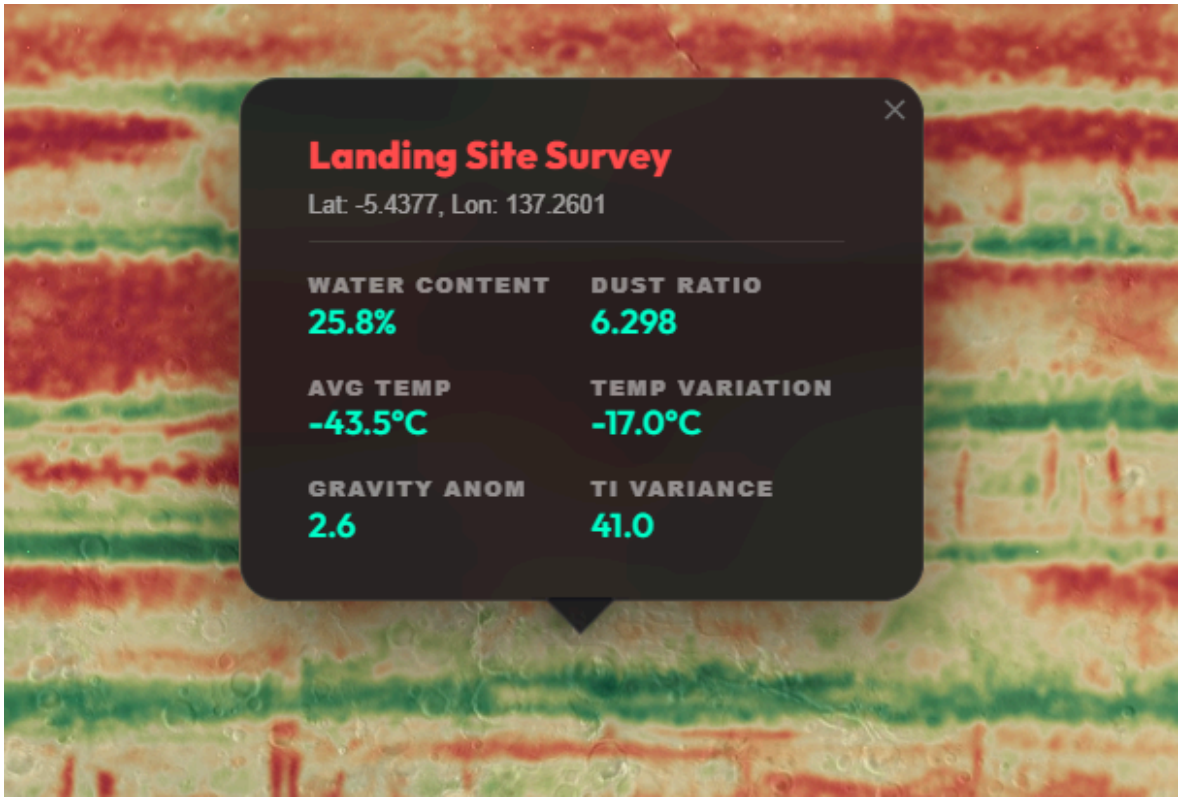
Weighting Effects

Adjusting target weights had clear effects on landing score distributions. Increasing the weight of Water prioritized regions with ice and crater structures, while applying higher Dust penalties shifted preference toward cleaner, less dusty terrain. Surface Temperature and Gravity weighting allowed fine-tuning for mission-specific needs, illustrating the flexibility of MGPF as a decision-support tool.

Table 4: *R^2, MAE, and Uncertainty of Targets*

Model	Target Variable	Test R^2	Test RMSE	Test MAE	Mean Uncertainty (σ)
Water	Mars Odyssey Neutron % WEH	0.9693	4.0600	1.7365	1.1206
Dust	OMEGA Ferric/Dust Ratio	0.8453	0.0087	0.0065	0.0019
Surface Temp	Yearly Avg Surface Temp (°C)	0.9976	0.9936	0.6073	0.2889
Temp Variation	Yearly Temp Variation (°C)	0.9808	2.3731	1.4312	0.5264
Gravity	Bouguer Gravity Anomaly	0.9097	55.3316	37.4577	17.8650
TI Variance	Thermal Inertia Variance	0.3416	24.9345	15.5019	3.3174

Figure 2: ***Heatmap predictions of Gale Crater on MGPF app***



According to Cermak (2012), Gale Crater was chosen for the Curiosity Rover for its potential habitability over its competitor site, Eberswalde. The heatmap indicates high water content near Gale Crater, consistent with its habitability criteria. The near-zero gravity anomaly (2.6 mGal) reflects the stable, low-density crater floor geology expected at the site.

Figure 3: ***Heatmap Predictions near Jezero Crater on MGPF app***



According to Nasa (2020), Jezero Crater was chosen "as the landing site for the Perseverance rover because scientists believe the area was once flooded with water and was home to an ancient river delta" (para. 1). Results on the heatmap indicate high water content near the crater, consistent with its history as an ancient lake basin. The negative gravity anomaly (-53.2 mGal) reflects low-density sedimentary infill characteristic of ancient lacustrine deposits, further supporting the site's geological significance.

Discussion and Conclusions

Discussion

The results from MGPF show that geophysical properties on Mars can be predicted with high accuracy using only surface-level features, even when the underlying datasets are noisy or low-resolution. Exceptional performance in targets like Surface Temperature and Temperature Variation suggests that certain geophysical variables have strong, learnable relationships with

surface features, while strong Water and Gravity predictions further demonstrate that the pipeline generalizes well across physically distinct target types. Weaker performance in TI Variance highlights the limitations of derived noisy targets rather than the pipeline itself, as the diurnal swing metric compresses physical information in ways that obscure learnable patterns. These outcomes indicate that networks can capture meaningful relationships between surface and geophysical characteristics across most target types, with data quality and target construction remaining the primary constraints on predictive ceiling.

These findings connect well with earlier work in Martian terrain analysis. Meng et al. (2024) demonstrated that convolutional neural networks can reliably identify geomorphological analogues between Earth and Mars, showing that surface morphology contains enough structure for ML models to learn from. Barrett et al. (2022) expanded on this idea by using NOAH-H to classify terrain hazards in HiRISE imagery, proving that deep learning can streamline data interpretation. MGPF adds to these contributions by showing that surface data can also be used to infer geophysical properties, which neither Meng et al. nor Barrett et al. attempted. This positions MGPF as a tool that builds on existing ML-based approaches while addressing a different part of the landing-site evaluation pipeline.

The heatmap validation further supports the usefulness of MGPF. High-scoring regions near Gale Crater and Jezero Crater aligned with the scientific reasoning behind their historical selection, suggesting that the framework captures the same broad patterns that mission planners have relied on. The ability to adjust target weights also makes the system flexible, allowing different missions to prioritize different scientific or engineering needs. Results indicate that MGPF can help narrow down candidate regions early in the site-selection process, reducing the amount of manual scanning required and making the workflow more efficient.

Conclusion

Overall, MGPF provides a scalable way to accelerate preliminary Martian landing site evaluation. By extrapolating high-resolution features from easily obtained surface data, it reduces the need for exhaustive scans. The system's weighted landing goodness heatmaps align with historical site selection and offer actionable insights for identifying candidate regions, adapting efficiently to changing mission priorities. Future work includes adding surface level data as adjustable predictors of landing suitability, as including variables like slope will make MGPF better suited to address engineering constraints as well as scientific ones.

Limitations

Data Quality

The highest available training quality is 2ppd due to WEH% having a low scan resolution. This decreases the total amount of data points that MGPF has to train on. Additionally, noise and inaccuracies in survey tools propagate into the model training despite heavy preprocessing and tuning.

The poor performance of models like TI variance can be attributed to this low data quality. High levels of noise obscure and shift the true relationships within the dataset, preventing the models from learning stable patterns. As a result, their predictive accuracy deteriorates significantly.

Speed

At 8ppd, data files are enormous and hard to load due to RAM issues. To mitigate the issue and allow for the heatmap to load, CSV has to be compressed to Parquet files, and even those have to

be lazy-loaded into RAM to prevent overload. This causes the API server to be slow and heatmap refreshes after zooming or adjusting weights to be delayed.

Hardware constraints also cap how much the models can improve. Limited device memory and processing power prevent certain leaps in training complexity, making it impractical to run more epochs or experiment with heavier hyperparameters. As a result, the models cannot fully converge or benefit from more advanced optimization strategies, which restricts their overall performance ceiling.

Scale

Currently, the model only features 6 targets, with 1 being too noisy for practical use. MGPF does not prioritize surface level targets like slope or roughness in the models, as they are easily obtainable for large areas at high resolutions compared to the targets. However, due to this, they are not implemented in the heatmap. This causes the heatmap to be more oriented toward scientific mission parameters like water density and dust, while excluding engineering restraints. MGPF's framework is built so that these limitations are easily addressable in the future. Adding new models to the framework is straightforward. The new model will have to be trained with the same paradigm as current models, and must be added to the extrapolation grid. Then, they must be added to the heatmap engine and a slider must be added. All of this is fairly easy to implement and allows MGPF to grow with new needs rather than a static tool that cannot be improved upon.

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